

<ProjectName>

Test Plan

Version:1.0

<DD/MM/YY>

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Security Classification: Confidential

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Document History

Document Location	
File Name	Location

Document Version History					
Version	Effective Date	Author	Details	Reviewer	Approvers

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1. Introduction

1.1 Purpose

This Test Plan aims to gather the information necessary to plan and control the test effort for the <ProjectName>. It describes the approach to testing the software and will be the top-level plan used by QCs to direct the test effort.

This Test Plan is covered the following objectives:

- Identifies the items that should be targeted by the tests.
- Identifies the motivation and ideas behind the test areas to be covered.
- Outlines the testing approach that will be used.
- List the deliverable elements of the test activities.

1.2 Glossary

Term	Description
PM	Project Manager
QC	Quality Controller
UI	User Interface
Smoke Test	A subset of all defined/planned test cases that cover the main functionality of a component or system to ascertain that the most crucial functions of a program work, but not bothering with finer details.
Test Case (TC)	A set of input values, execution preconditions, expected results, and execution postconditions, is developed for a particular objective or test condition, such as to exercise a particular program path or to verify compliance with a specific requirement.
Test Objective	The purpose for designing and executing a test or a test type.
Test Schedule	A list of activities, tasks, or events of the test process, identifying their intended start and finish dates and/or times and interdependencies.

1.3 Project Context and Background information

TBD

1.4 Document Scope

1.4.1 In Scope

The software development process involves various types of testing. Each test type addresses a specific testing requirement. As discussed with CLIENT, the following black box testing types will be applied in the testing phase of the <ProjectName>:

1.4.1.1 UCs

As the list of UCs might change (slightly via time), please refer to the content in the high-level sheet schedule in the project weekly RAG report starting from 2020 02 07.

1.4.1.2 Testing types

No.	Testing Type	Web - Application	Mobile Application	Remarks
1	Smoke testing	YES	YES	Web application: Windows 10 – Chrome, MS Edge Legacy Mobile application: iPhone 11 Pro and Samsung S10
2	Manual Functional Testing	YES	YES	Based on the use cases
3	Manual API Testing	YES	YES	Using the free version of Postman to test the public API
4	Cross-Browser Testing	YES	NO	Firefox in Windows 10 and Safari in MAC
5	Compatibility Testing	NO	NO	
6	User Interface Testing	YES	YES	
7	Localization Testing	YES	YES	
8	End2End Testing	YES	YES	
9	User Acceptance Testing	YES	YES	
10	Performance Testing - API	YES	YES/NO	- For web application: included in #13 “Performance Testing - End 2End (UI).” - For Mobile native application: In the case in the design or implementation if there are any APIs for Mobile only, NashTech must perform this test type for Mobile App
11	Partial Regression Testing	YES	YES	This test type will be done after verifying bugs

12	Complete Regression Testing	YES	YES	
13	Performance Testing - End 2End (UI)	YES	NO	Measure the response time of 20 end2end scenarios for both front-end and back-end. Each scenario has less than 10 steps in maximum and without import/export functions. The number of concurrent users will be distributed in all scenarios. Target of response Time: <=3s for ~100 concurrent users and <=5s for ~5000 concurrent users
14	Security Testing	YES	YES	"The security testing will be done two times in the 2nd release and the last release. It will be tested in a separate testing environment. The first is quickly checking the security risk without a detailed report."
15	Automated Testing - API	NO	NO	
16	Automated testing - UI	NO	NO	
17	Stress Testing	NO	NO	
18	Responsive Testing	YES	NO	For this phase, responsive testing will be checked on: 1. iPhone 11 pro and Samsung S10 for mobile native application as below: When the user goes to the web: - If the device has not installed the mobile app yet, then an information page shows how to install the mobile native application - Else mobile native app is opened 2. iPad Pro for a web application: - it displays the same desktop - TCs set of the smoke test is picked to run on a simulator environment instead of the real device.

1.4.1.3 Testing Environment and Testing Type

	Smoke Testing	Functional Testing	UI Testing	Cross Browser Testing	Regression Testing	Performance Testing (*)	Security Testing (*)	End2End Testing	Localization Testing	Manual API Testing (*)	Responsive Testing
Win 10 – MS Edge Legacy	Yes	No	No	No	No	No	No	No	No	No	Refer to 1.4.1.2 item 18 above
Win 10 – Chrome The latest version at the testing time	Yes	Yes	Yes	No	Yes	No	No	Yes	Yes	No	
Win 10 – Firefox The latest version at the testing time	No	No	Yes	Yes	No	No	No	No	No	No	
Mac - Safari	No	No	Yes	Yes	No	No	No	No	No	No	
Iphone 11 Pro(**)	Yes	Yes	Yes	No	Yes	No	No	Yes	Yes	No	
SamSung S10(**)	Yes	Yes	Yes	No	Yes	No	No	Yes	Yes	No	

(*) The test types are not dependent on the browser and will be tested in a separate testing environment, so they are marked “NO” in this section.

(**) iPhone 11 Pro and Samsung S10 are used for testing the functionality of Mobile applications.

1.4.2 Out of Scope

The following requirements are not in the scope of the testing phase:

Requirement	Reasons
Compatibility Testing	The system will test in specific browsers and devices as per the agreement.
Stress Testing	This testing type should be considered to estimate and implement in the second phase (after 15-May-2020)
Automated Testing – API	This testing type should be considered to estimate and implement in the second phase (after 15-May-2020)
Automated testing - UI	This testing type should be considered to estimate and implement in the second phase (after 15-May-2020)
Pixel Perfect for UI/UX	There is no specific requirement for this kind of test.

Requirement	Reasons
Data Migration Test	There is no database migration. NashTech is to provide a minimum dataset to initial the application.

1.5 Constraints

In this project, there are the following constraints:

- Development is delayed. It will affect the progress of the testing.
- More defects are found, which require more time to verify and fix.
- Test types not mentioned in the test plan will not be tested.

1.6 Risk List

A list of risks that may affect the design, development, or implementation of testing.

No.	Risk	Mitigation Strategy	Contingency (Risk is realized)
1	Review TCs is delayed by CLIENT.	QC team still executes TCs before and considers updating them later when CLIENT feedback.	New TCs can be tested as soon as or delayed; this depends on the complexity of TCs.
2	Review TCs's feedback doesn't belong to the current scope	BA needs to reconfirm with CLIENT and then create enhancement	PM and BA should define the process and clear this point to the Client to avoid this issue.
3	Using dummy data for testing so the issue can happen with real data	It will be better if client can provide example data that is the same as real data.	Store data that lead to bugs to use in the next testing
4	CLIENT can't provide the 3 rd party information on time: ▪ QC team can't do integration testing between 3rd party and our system ▪ Functions cannot be delivered on time	Request clients provide 3 rd party information before a function is implemented.	The development team needs to raise this with the client and PM soon.
5	The UI layout is not fully covered the information in the use case. It may have inconsistency information between the UI layout design and application.	QC team is based on the information in the use case for testing purposes.	
6	Build delivery is late from the development team. QC team can: ▪ Miss and slip test plan due to late build delivery from the developer team. ▪ Not complete tasks on the test plan.		▪ QC team must re-plan the Test Plan according to a real case of late build delivery. ▪ QC team works OT because the plan is slipped into comparison with a plan before.

No.	Risk	Mitigation Strategy	Contingency (Risk is realized)
	Work OT or reduce testing scope in a plan to meet the deadline.		

2. Test Approach

2.1 Test Schedule

The test schedule is based on the project plan, which is managed by the project manager

2.2 Testing Types

2.2.1 Smoke Testing

The smoke test should be thorough enough that if the build passes, QC can assume it is stable enough to be tested more thoroughly. Smoke testing is done by QCs before accepting a build for further testing of this build.

Test Objective	Smoke testing is the initial testing process; we check that the system under test works, returns normally, and does not blow up. This testing results to decide if a build is stable enough to proceed with further testing.
Technique	- A set of test cases that covered the basic flow will be marked "YES" in the "smoke" column of the complete function TCs. - This set of TCs will be used to execute the smoke testing.
Completion Criteria	All TCs marked "YES" in the "smoke" column have been executed.
Special Considerations	- This test will be done on Windows 10 – Chrome for web application - This test will be done on iPhone 11 Pro and Samsung S10 for mobile native application - The release package will be rejected if the package has one of the following issues: - Cannot execute testing - Main functions do not work

2.2.2 Manual Functional Testing

Function testing of the target-of-test should focus on any requirements for a test that can be traced directly to use cases or business functions and business rules. This test type verifies that the system works as required and validates that the correct functionality has been delivered.

Test Objective	Verify proper data acceptance, processing, and retrieval and the appropriate implementation of the business rules mentioned in each use case.
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Technique	▪ Create the test cases to cover all business rules and workflow in each use case. ▪ Using the boundary testing technique to verify the following: – The expected results occur when valid data is used. – The appropriate error or warning messages are displayed when invalid data is used.
Completion Criteria	▪ All test cases have been executed.
Special Considerations	▪ Each business rule is properly applied. ▪ This test will be done on Windows 10 – Chrome for web application ▪ This test will be done on iPhone 11 Pro and Samsung S10 for mobile native application

2.2.3 User Interface Testing

User Interface (UI) testing verifies a user's interaction with the software. The goal of UI testing is to ensure that the User Interface provides the user with the appropriate access and navigation through the functions of the target-of-test.

Test Objective	The objective of this testing type is to: ▪ Verify that all UI requirements are the same as mentioned in UCs. ▪ All forms are consistent and do not break.
Technique	▪ UI TCs will be included in the manual functional TCs
Completion Criteria	▪ Each form was successfully verified by using UI cases in the manual functional TCs
Special Considerations	The test case for UI testing is included in each manual functional test case.

2.2.4 Cross Browser Testing

This type of test will focus on using different web browsers to access the system and make sure that there is no difference in the GUI and that the functionality works correctly across web browsers.

Test Objective	Verifying: ▪ Main functions work correctly in all selected browsers ▪ All UI of form are the same as mentioned in UCs.
Technique	▪ The TCs belong to the main flow of UCs, and UI cases are marked "YES" in the "Cross Browser" column of the manual functional TCs
Completion Criteria	▪ All TCs marked "YES" in the "Cross-browser" column have been executed.
Special Considerations	This testing is only performed once features/UIs have been stabilized in the primary browser. This test will be done in the following environment:

- Windows 10 - Firefox
- iMAC – Safari

2.2.5 Regression Testing

Regression Testing is to re-execute the test cases to check whether the previous functionality of the application is working fine and the new changes have not introduced any new bugs.

In this project, there are two kinds of Regression testing types:

- Partial Regression Testing (Confirmation Testing): This is regression testing, and the scope of this test is related to the bug-fixing of the functionality or the enhancement of the existing feature in the new build.
- Complete Regression Testing: This is regression testing, and the test's scope is related to all system functionality. The team will plan to do this kind of regression test 1-2 weeks before the final release.

2.2.5.1 Partial Regression Testing

Test Objective	Partial Regression testing is to verify that a code change (fixed bug, add/modify functionalities) in the software does not impact the existing functionality.
Technique	▪ The team is to base on the bug description and the build note to identify which feature needs to be done for this test.
Completion Criteria	▪ All selected test cases are executed.
Special Considerations	▪ Depending on the build plan and bug fixing, the selected test cases, and test browsers will be adjusted. ▪ This kind of test will not be recorded.

2.2.5.2 Complete Regression Testing

Test Objective	Complete Regression Testing is to re-execute executed to test the complete product end to end by covering all the core functionalities of the Product.
Technique	▪ The team is to select <u>about</u> 30% of Manual Functional Test cases for doing this kind of test
Completion Criteria	▪ All selected test cases are executed.
Special Considerations	This test will be done 1-2 weeks before the release date.

2.2.6 Responsive testing

Responsive testing is used to check the appearance of a website with a responsive web design. This design helps a website render the content in a flexible yet appropriate manner on the basis of the OS on which it gets summoned.

Test Objective	Verify that: - UI requirements are the same as mentioned in UCs on iPadPro - Forms are consistent and do not break.
Technique	TCs set of the smoke test is picked to run on a simulator environment instead of the real device.
Completion Criteria	All TCs marked "YES" in the smoke column are run.
Special Considerations	The responsive test will be done on a simulator environment for iPadPro.

2.2.7 End2End Testing

End-to-end testing is the process of testing transactions or business-level products as they pass right through the computer systems. Thus, this generally ensures that all aspects of the business are supported by the systems under test.

E2E testing schedule is based on the project plan, which is managed by the project manager.

Test Objective	Ensures that the overall process flows as expected, that is, that system components integrate together correctly and that the right information is passed between them
Technique	Create scenarios to go through happy cases of all functions in the project scope.
Completion Criteria	Scenarios cover almost of business operations in the project scope.
Special Considerations	- This test will be run in the latest build with a client environment. - This test will be done on Windows 10 – Chrome

2.2.8 Localization Testing

Localization testing verifies how well the build has been translated into a target language.

Test Objective	Is to verify that the system supports multi-languages
Technique	- Update the language files to another language, such as Vietnamese or English. - Run the end2end test cases to go through all functions of the system
Completion Criteria	All functions have been executed.
Special Considerations	- The test will be run in the latest build. - There is no verification of the corrective of the language. The testing is to verify that there is no hard code.

2.2.9 APIs Testing

API (Application Programming Interfaces) testing is different from other software testing types as GUI is unavailable. Instead of standard user inputs and outputs, API testing uses software to send/call to the API and get the output.

Test Objective	Testing APIs occurs at the business logic layer of the software architecture. It is to verify that the output meets the acceptance criteria defined in the user story.
Technique	- Using the Boundary Value Analysis (BVA) and Equivalence Partitioning (EP) techniques to create the test data for each API and verify its response. - Tools: Postman
Completion Criteria	All public APIs are tested.
Special Considerations	Just test the public API of the system.

2.2.10 Performance Testing

The goal of the Performance test is to verify that performance requirements have been achieved in a specific environment and configuration of the system. For this project, the requirement is to have all pages loaded within one second under the condition that there are 1000 concurrent users accessing the system.

Test Objective	Measure the response time of web application at the end user level to prove whether the system's performance is met performance requirements. - Run 10 end2end scenarios in the front end. It is to measure the response time of the system - Run 10 end2end scenarios in the back-end office. It measures the performance when they change the setting (different configuration and configuration changes). Target Response Time: $\leq 3s$ for 100 concurrent users and $\leq 5s$ for 5000 concurrent users in the normal internet traffic The test will be run with 5000 concurrent users, and the number of concurrent users will be distributed in pages in the system.
Technique	- According to the function list, the QC will create the performance test scripts using the Jmeter. - The Blaze Meter will be used to generate the concurrent users. - The test will use the ratio concurrent users, which is defined in the "Appendix – Performance Testing - Use case and loading ratio" section, and it is repeated by following the defined number of users: - Ram up 5 users every 1 second, until 100 concurrent users - Loading 10 minutes - Ram down and stop.

	– Analyze results and generate test reports. – Repeat this test with 5000 concurrent users
Completion Criteria	▪ All planned test scripts have been executed. ▪ Report about the response time, CPU, and RAM in use as defined in the Key Performance Indicators (KPI) Monitoring ▪ Highlight the abnormal response and work with the development team to find out the root cause and improvement point, if any.
Special Considerations	▪ Don't do the performance test for the function related to 3rd party services & import/export functions ▪ Database for performance testing: There are 1000 properties in the system. NashTech will input the dummy data for the testing. According to the result (response time, CPU in use, RAM in use) of the first performance testing time with 5000 concurrent users, NashTech will raise to CLIENT to consider on upgrade the hardware of the production environment if needed. The target of response time is ideal because the response time depends on the internet bandwidth. Web applications and mobile applications use the same APIs. The performance test in the web application will cover the APIs.

2.2.11 Security Testing

The goal of security testing is to identify the threats in the system and measure its potential vulnerabilities, so the system does not stop functioning or is exploited. It also helps detect all possible security risks in the system and helps developers fix these problems through coding.

Test Objective	Verify the security risks of Web Applications and mobile applications based on the OWASP Top 10 standard Mobile and web applications penetration testing can be verified against OWASP TOP 10 Security level 2 and Web Application Security Requirements 4.0 based on generally recommended for this project.
The target of the test	▪ Web Application ▪ Mobile Application (android and iOS)
Types of testing	▪ Black box testing ▪ Grey box testing
Completion Criteria	▪ NashTech provides a comprehensive web application security report that's understandable by the dev team. ▪ Only failed test cases are provided with details like a step to reproduce, screenshots
Special Considerations	▪ The web application security test will be performed at when the end of development. ▪ The penetration testing will not test areas/features of 3rd party libraries or components ▪ None of the detail test cases is for this kind of test. ▪ This will be tested by the security testing team.

2.2.12 User Acceptance Testing (UAT)

User Acceptance Testing aims to assess if the system can support day-to-day business and user scenarios and ensure the system is sufficient and correct for business usage.

This test plan is not covering the UAT. The UAT plan will manage in a separate document by the project manager.

QC team still supports the UAT period by verifying the defect and hotfix of the system.

3. Resources

3.1 Human Resources

This table shows the staffing assumptions for the project.

Role	Resource Recommended	Responsibilities / Comments
QC Lead	1	Responsible for all testing activities mentioned in this document, including the following: - Create a test plan - Review test cases - Deploy the release package to the test environment. - Report the testing progress to PM and QCM - Support to analyze the change requests - Ensure all test methods are qualified & verified after transferring and verified as required. - Managing and monitoring all QC members' tasks - Communicate with other functions to align multifunctional operations and maintain the quality of the project. - Play the QC role in the project
SQC	2	The same QC lead excludes creating test plan
QC	2	- Create Test cases/Test scripts - Execute test - log results - Analyze and recover from test failures
DEV Team Lead		Analyze and categorize the bugs and follow up to make sure that the bug is fixed.
Project Management		- Provide the documents and monitor the project progress. - Review/Approve the test plan, test report, and final summary report.
QC Manager		- Review the Test plan. - Handles escalations from senior management and the test team

3.2 Test Environment

The development and testing environment are separated. Only the QC team has permission in the test environment. The system under test will be hosted in the cloud.

3.2.1 Hardware

Use standard NashTech Development machine provided by IT department.

3.2.2 Software

Device	Resolution	System	Browser	Note
PC	1366x768	Windows 10	Chrome, Firefox	All browsers will be in the latest version. (The version will be updated when the team starts testing)
iMac	1366x768	IOS 10.15.1	Safari	For testing the web application
iPhone 11 Pro	default	IOS 13.1.3		For testing the native iOS application, only
Samsung S10	default	Android 9		For testing the native Android application, only

4. Deliverables

The list of deliverables in the testing phase is defined in the RAG report.

5. Measurement

No	Quality Indicator	Responsibility	Frequency
1	Defect Density	QCL / SQC	Weekly
2	Defect leakage	QCL / SQC	Weekly – in the UAT period
3	Defect Removal Efficiency	QCL / SQC	Weekly – in the UAT period
4	Invalid bugs rate	QCL / SQC	Weekly – in the UAT period

6. Suspension / Exit Criteria

6.1 Suspension Criteria

If any defects are found that seriously impact the test progress, the test team may choose to

Suspend testing. Criteria that will justify test suspension are:

- Hardware/software is unavailable at the time indicated in the project schedule.

- Source code contains one or more critical defects, which seriously prevents or limits testing progress.
- Assigned test resources are not available when needed by the test team.
- Main functions in the release note had not been properly met.

6.2 Resumption Criteria

If testing is suspended, resumption will only occur when the problem(s) that caused the suspension has been resolved. When a critical defect is the cause of the suspension, the “FIX” must be verified by the test department before testing is resumed.

6.3 Exit Criteria

Testing will be considered complete when the following conditions have been met:

- All test cases in all areas have been executed.
- All priority 1 and 2 bugs have been resolved and closed.
- No defects must be left in an open, unresolved status.
- The outstanding bugs are less than the number of bugs that are agreed upon for entry UAT

Severity	Description	Max Number Allowed	
		For entry into UAT	For exit out of UAT
Critical	▪ Showstopper - An error that makes the whole or a significant part of the application inoperable or otherwise has a significant effect on Customer's business. ▪ The module/product crashes or the bug causes non-recoverable conditions, database or file corruption, the program hangs requiring a reboot ▪ Application or service does not work as intended; impossible to use the application or a main function of the application; critical functionality lost ▪ Blocks development and/or testing work (new build, with a fix, needed immediately) e.g., system crash	0 Showstopper - QC cannot proceed with any tests	0
Major	▪ An error that has a material effect on the functionality, performance, or user experience of the application function upon which the Customer relies for the efficient conduct of its business. ▪ Major system component unusable due to failure or incorrect functionality, prevents other areas of the app from being ▪ Operation is significantly impacted; User is possibly led to creating future problems.	0 QC cannot proceed with any tests in a functional area	0

Severity	Description	Max Number Allowed	
		For entry into UAT	For exit out of UAT
	Can have a workaround, but the workaround is inconvenient or difficult		
Normal	Spelling or grammatical mistakes in the User Interface Unfriendly behavior that is hindering but workable, for the user or service There is a simple workaround for the bug	10 It does not stop testing - but limits it e.g., data validation wrong	5
Minor	A minor or cosmetic error that that causes a minimal effect on the Customer's business	20	10

7. Appendix

7.1 Appendix – Performance Testing - Use case and loading ratio

Loading ratio means % of the user of total users will be distributed in the scenario.

No.	Category	Feature	Scenario	Loading ratio
1	Front-office	Feature 1	TBD	15%
2	Front-office	Feature 2	TBD	5%
3	Front-office	Feature 3	TBD	15%
4	Back-office	Feature 4	TBD	5%
5	Back-office	Feature 5	TBD	20%
6	Front-office	Feature 6	TBD	20%
7	Front-office	Feature 7	TBD	5%
8	Front-office	Feature 8	TBD	5%
9	Back-office	Feature 9	TBD	5%
10	Front-office	Feature 10	TBD	5%

7.2 Key Performance Indicators (KPI) Monitoring

An application under test: The below metrics will be captured and used to report per execution

7.2.1 UI Response Times:

- Min
- Median
- Average
- Maximum
- 90%
- Error rate

7.2.2 Application & Database Server - Azure Monitoring

- Memory & CPU: Monitor Metrics

8. Reference and Related Documents

No.	Document	Comments	Version
1	SRS documents		1.1
2	WBS		1.8

We are experts in technology, delivering smart solutions that solve business challenges and create value. Our award-winning teams apply deep expertise and passion to deliver complex IT projects globally.

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